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Informatica® Cloud Application Integration
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NVIDIA NIM Connector Guide

Informatica Cloud Application Integration NVIDIA NIM Connector Guide
May 2025

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Table of Contents

Preface 5

Chapter 1: Introduction to NVIDIA NIM Connector 6

NVIDIA NIM overview 6

Chapter 2: NVIDIA NIM connections 8

Basic connection properties 8

NVIDIA NIM connection properties 9

Create an NVIDIA NIM API key 9

NVIDIA NIM connection metadata 9

NVIDIA NIM connection actions and objects 10

Using an NVIDIA NIM connection in a process 23

Preface

Read the *NVIDIA NIM Connector Guide* to learn how to set up and use NVIDIA NIM connections. Use an NVIDIA NIM connection to leverage the power of NVIDIA's accelerated computing to access and run a variety of AI models. This guide assumes that you have an understanding of NVIDIA NIM concepts.

CHAPTER 1

Introduction to NVIDIA NIM Connector

The NVIDIA NIM Connector is an integration tool that connects NVIDIA's NIM (NVIDIA AI Enterprise Management) platform with various data and analytics systems. It enables seamless management and orchestration of AI workloads on NVIDIA's GPU-accelerated infrastructure, helping organizations to optimize resource utilization, streamline AI workflows, and accelerate machine learning and deep learning initiatives.

NVIDIA NIM is a set of easy-to-use microservices designed for secure, reliable deployment of high-performance AI model inferencing across clouds, data centers and workstations. The connector enables users to easily integrate any NVIDIA NIM into their agentic process orchestration workflows.

By leveraging NVIDIA NIM through IDMC, businesses can skip complex setup processes and instantly access production-ready AI, ensuring high performance, scalability, and access to the latest NVIDIA innovations without requiring extensive technical expertise.

NVIDIA NIM Connector provides the following key features:

- Easily integrates advanced language understanding and generation for tasks, such as intelligent customer service bots, automated report summaries, and complex decision support, with features such as chat completion, function calling, and structured output.
- Creates accurate, context-aware knowledge discovery systems using advanced embeddings and reranking models, enhancing internal search and Retrieval-Augmented Generation (RAG) applications.
- Extract deeper insights from unstructured data by integrating capabilities, such as image analysis, document understanding with Optical Character Recognition (OCR), and image generation into business processes, such as quality control and asset management.
- Deploys pre-optimized, domain-specific models for industries, such as healthcare diagnostics or climate simulations, leveraging NVIDIA's domain-specific expertise.

NVIDIA NIM overview

NVIDIA NIM offers a cloud-native deployment model, allowing users to run models using NVIDIA API endpoints or in their own environment using NGC containers. It supports multi-model and multi-instance serving, efficiently utilizing resources by hosting multiple models concurrently with batching, and scaling across single or multiple GPUs. The flexible hosting options include on-premises deployment with NGC

containers, NVIDIA cloud-hosted endpoints, and localhost inference endpoints commonly used for development.

NVIDIA NIM covers a broad range of use cases, supporting natural language processing tasks such as text generation, reranking, embeddings, summarization, classification, and translation. It also supports vision tasks including image classification, segmentation, and object detection.

CHAPTER 2

NVIDIA NIM connections

Use NVIDIA NIM Connector to leverage the power of NVIDIA's accelerated computing to access and run a variety of AI models. Services include basic inference and advanced tasks, such as healthcare model invocations, climate simulations, medical imaging analysis, and multimodal data processing.

You can perform actions, such as get polling status, invoke basic model, invoke climate simulation model, invoke healthcare model, invoke medical imaging model, and invoke multimodal model.

After you create an NVIDIA NIM connection, you can validate and save the connection.

You can then publish the NVIDIA NIM connection and click the **Metadata** tab to view the generated process objects for the connection.

Basic connection properties

The following table describes the basic properties that you can configure on the **Properties** tab of the connection creation page:

Property	Description
Name	Required. Unique name for the NVIDIA NIM connection that identifies it in the Process Designer. The name must start with an alphabet and can contain only alphabets, numbers, and hyphens (-).
Location	The location of the project or folder where you want to save the connection. Click **Browse** to select a location. If the **Explore** page is currently active and a project or folder is selected, the default location for the connection is the selected project or folder. Otherwise, the default location is the location of the most recently saved asset.
Description	Optional. Description of the connection.
Type	Required. The type of connection you want to use for the connector or service connector. Select **NVIDIA NIM**.
Runtime Environment	Required. The runtime environment for the connection. You can run the connection on the Cloud Server, a Secure Agent group, or a Secure Agent machine.
Connection Test	Not supported for NVIDIA NIM Connector.
OData-Enabled	Not supported for NVIDIA NIM Connector.

NVIDIA NIM connection properties

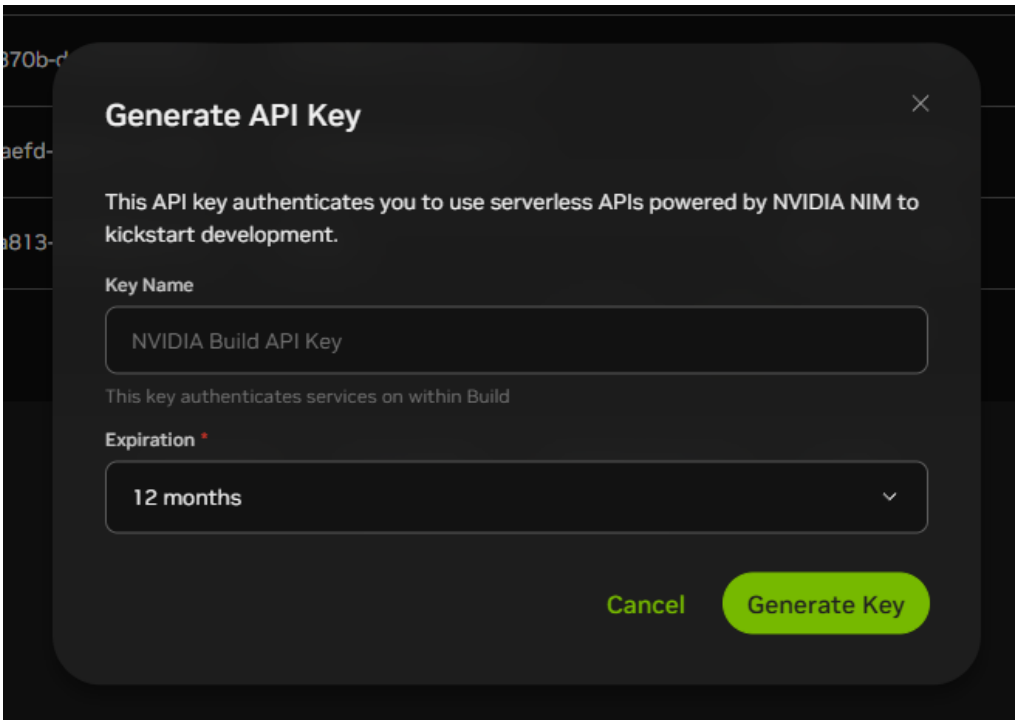
To use the NVIDIA NIM connection, you need the API key. Enter the API key in the **Api_Key** property in the **Connection Properties** section. The API key is used to authenticate NVIDIA NIM API requests.

Create an NVIDIA NIM API key

Create an NVIDIA NIM API key for authentication to use the NVIDIA NIM APIs.

1. Log in to the NVIDIA API Catalog.
2. Go to **Settings** > **API Keys**.
3. Select the duration for which you want the key to remain activated from the **Expiration** list and click **Generate Key**.

   The following image shows the **Generate Key** option on the Generate API Key dialog box:



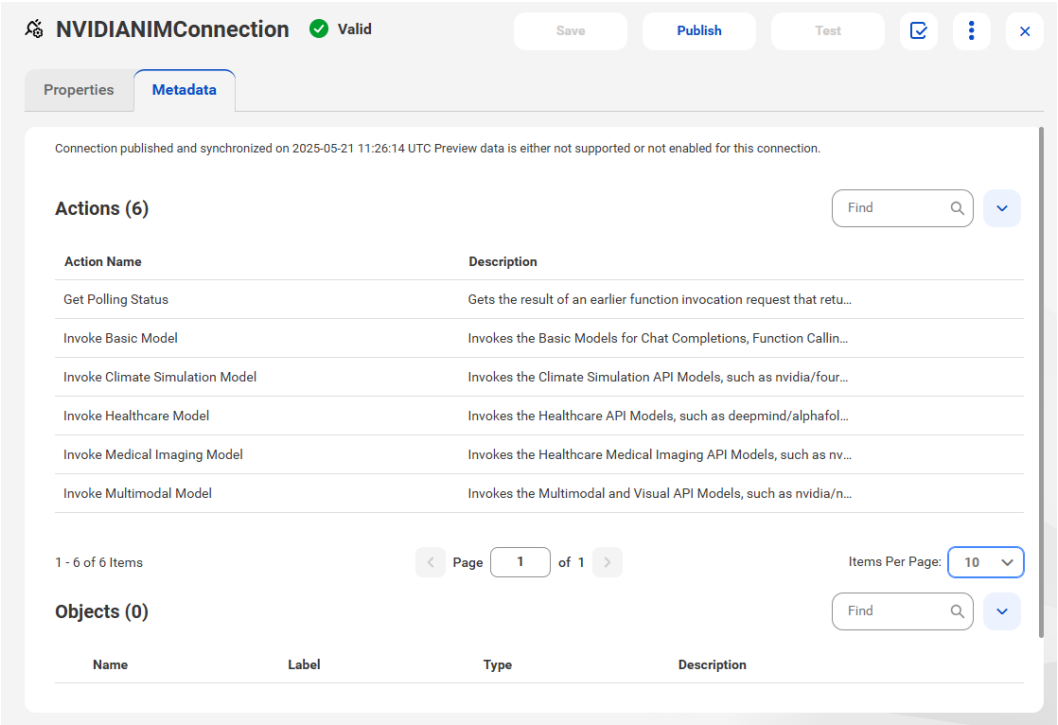
4. Copy the key and store it in a secure place.

NVIDIA NIM connection metadata

After you create an NVIDIA NIM connection, you can validate and save the connection. You can then publish the NVIDIA NIM connection and click the **Metadata** tab to view the generated process objects for the connection.

When you publish an NVIDIA NIM connection in Application Integration, the **Actions** and **Objects** appear on the **Metadata** tab.

The following image shows the **Metadata** tab for a published NVIDIA NIM connection:



NVIDIA NIM connection actions and objects

When you publish an NVIDIA NIM connection in Application Integration, the **Actions** and **Objects** appear on the **Metadata** tab.

You must escape special characters in payloads as shown in the following table:

Symbol	Character	Replacement
<	left angle bracket or less than	<
>	right anlge bracket or greater than	>
{	left curly bracket or opening brace	{{
}	right curly bracket or closing brace	}}
&	ampersand	&

To use fields with data types, such as array, boolean, double, or integer, you must use annotation as shown in the following sample code:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<field_array m:isArray="true">
<property_1> ...</property_1>
</field_array>
<field_boolean m:type="xs:boolean">true</field_boolean>
<field_double m:type="xs:double">10.3</field_double>
```

```
<field_integer m:type="xs:int">5</field_integer>
</root>
```

Important:

- Creating assets and referencing them in payloads is not supported in the current version of the connector.
- Some models might return responses larger than 5 MB. It is recommended to publish the connection on an agent to handle large files efficiently.

Consider the following information when you use specific actions and objects:

- The **Invoke Healthcare Model** action invokes the Healthcare API models, such as deepmind/alphafold2 and nvidia/genmol. It invokes NVIDIA healthcare models to generate predictions or analysis based on the input data.
  The connector is tested with the following models:

  - Deepmind/alphafold2
  - Nvidia/esmfold
  - Proteinmpnn/predict
  - Nvidia/genmol
  - Nvidia/molmim

  You must provide values to the input fields, such as payload, version, and NVCF_POLL_SECONDS.

  For example, consider the nvidia/genmol model with the following endpoint URL:

  `https://health.api.nvidia.com/v1/biology/nvidia/genmol/generate`

  In this URL, `v1` is the API version and `biology`, `nvidia`, `genmol`, and `generate` are the individual URL path segments.

  NVCF_POLL_SECONDS - interval in seconds between status checks for a pending request to an NVIDIA Cloud Function (NVCF). The maximum supported value is 59 seconds.

  The following snippet is a payload example:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<smiles>C124CN3C1.S3(=O)(=O)CC.C4C#N.[*{20-20}]</smiles>
<num_molecules m:type="xs:int">30</num_molecules>
<temperature>0.01</temperature>
<noise>1</noise>
<step_size m:type="xs:int">1</step_size>
<scoring>QED</scoring>
<unique m:type="xs:boolean">false</unique>
</root>
```

  The following snippet is a sample response:

```
200 OK
<root>
  <molecules>
   <score>0.851</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(c2ccc(C(F)(F)F)cc2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.851</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(c2ccc(C(F)(F)F)cc2)C1</smiles>
  </molecules>
  ...
  ...
  <molecules>
   <score>0.851</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(c2ccc(C(F)(F)F)cc2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.851</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(c2ccc(C(F)(F)F)cc2)C1</smiles>
```

```
  </molecules>
  <molecules>
   <score>0.805</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(NC(=O)C2(C)CCC2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.802</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(OCc2ccc(Cl)cc2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.798</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(NC(=O)CCC(C)(C)C)C1</smiles>
  </molecules>
  <molecules>
   <score>0.785</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(OCC2CCCC(O)C2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.78</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(NC(=O)CCC(F)(F)F)C1</smiles>
  </molecules>
  <molecules>
   <score>0.78</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(NC(=O)CCC(F)(F)F)C1</smiles>
  </molecules>
    ...
    ...
  <molecules>
   <score>0.78</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(NC(=O)CCC(F)(F)F)C1</smiles>
  </molecules>
  <molecules>
   <score>0.773</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(NC(=O)C2CCCCCC2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.752</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(NC(=O)CCC(C)(F)F)C1</smiles>
  </molecules>
  <molecules>
   <score>0.75</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(C(=O)CCC2(O)CCC2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.668</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(C(=O)CCC2CCCCCC2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.655</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(OC(=O)CCC2CC2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.627</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(C(=O)CCCN2CCOCC2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.562</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(OC(=O)C2CC=CCC2)C1</smiles>
  </molecules>
  <molecules>
   <score>0.481</score>
   <smiles>CCS(=O)(=O)N1CC(CC#N)(OC(=O)CCCCC(=O)O)C1</smiles>
  </molecules>
  <status>success</status>
</root>
```

- The **Get Polling Status** action gets the result of an earlier function invocation request that returned a 202 status. It retrieves the result of a previous function invocation that returned a 202 status code.

For example, when invoking a model, if the response takes longer than the configured response timeout, the server returns a status header as in-progress and a status code 202. It also provides a request ID in the response header, which can be used to poll for the final result.

You must provide values to the input fields, such as host, version, and NVCF_POLL_SECONDS.

By default, the status polling is performed using the following URL:

```
https://optimize.api.nvidia.com/v1/status/{requestId}
```

In this URL, `optimize.api.nvidia.com` is the host, `v1` is the API version, and `status` is a URL path segment.

Note: Some models might use different status polling URLs. For more information, see the NVIDIA documentation for model-specific details.

The following snippets are some additional payload examples for other models:

- **Deepmind/alphafold2**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<sequence>MVPSAGQLALFALGIVLAACQALENSTSPLSADPPVAAAVVSHFNDCPDSHTQFCFHGTCRFLVQEDKPACVCHS
GYVGARCEHADLLAVVAASQKKQAITALVVVSIVALAVLIITCVLIHCCQVRKHCEWCRALICRHEKPSALLKGRTACCHSETVV
</sequence>
<algorithm>mmseqs2</algorithm>
<databases m:isArray="true">uniref90</databases>
<databases m:isArray="true">small_bfd</databases>
<e_value m:type="xs:double">0.0001</e_value>
<iterations m:type="xs:int">1</iterations>
<relax_prediction m:type="xs:boolean">false</relax_prediction>
<skip_template_search m:type="xs:boolean">true</skip_template_search>
</root>
```

- **Nvidia/esmfold**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<sequence>MVPSAGQLALFALGIVLAACQALENSTSPLSADPPVAAAVVSHFNDCPDSHTQFCFHGTCRFLVQEDKPACVCHS
GYVGARCEHADLLAVVAASQKKQAITALVVVSIVALAVLIITCVLIHCCQVRKHCEWCRALICRHEKPSALLKGRTACCHSETVV
</sequence>
<algorithm>mmseqs2</algorithm>
<databases m:isArray="true">uniref90</databases>
<databases m:isArray="true">small_bfd</databases>
<e_value m:type="xs:double">0.0001</e_value>
<iterations m:type="xs:int">1</iterations>
<relax_prediction m:type="xs:boolean">false</relax_prediction>
<skip_template_search m:type="xs:boolean">true</skip_template_search>
</root>
```

- **Proteinmpnn/predict**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<input_pdb> ATOM      1  N   GLY A   1      11.104  13.207  10.678  1.00
20.00           N
ATOM      2  CA  GLY A   1      12.560  13.207  10.678  1.00 20.00           C
ATOM      3  C   GLY A   1      13.104  14.207  11.678  1.00 20.00           C
ATOM      4  O   GLY A   1      12.560  15.207  11.678  1.00 20.00           O
TER
END </input_pdb>
<input_pdb_chains m:isArray="true">A</input_pdb_chains>
<ca_only m:type="xs:boolean">false</ca_only>
<use_soluble_model m:type="xs:boolean">true</use_soluble_model>
<random_seed m:type="xs:int">42</random_seed>
<num_seq_per_target m:type="xs:int">1</num_seq_per_target>
<sampling_temp m:isArray="true" m:type="xs:double">0.1</sampling_temp>
</root>
```

- **Nvidia/molmim**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<algorithm>CMA-ES</algorithm>
<smi>[H][C@@]12Cc3c[nH]c4cccc(C1=C[C@H](NC(=O)N(CC)CC)CN2C)c34</smi>
<num_molecules m:type="xs:int">30</num_molecules>
```

```
<iterations m:type="xs:int">10</iterations>
<property_name>QED</property_name>
<particles m:type="xs:int">30</particles>
<minimize m:type="xs:boolean">false</minimize>
<min_similarity m:type="xs:double">0.3</min_similarity>
<scaled_radius m:type="xs:double">1</scaled_radius>
</root>
```

- **nvidia/maisi**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
 <num_output_samples m:type="xs:int">1</num_output_samples>
 <body_region m:isArray="true">abdomen</body_region>
 <anatomy_list m:isArray="true">liver</anatomy_list>
 <output_size m:isArray="true" m:type="xs:int">256</output_size>
 <output_size m:isArray="true" m:type="xs:int">256</output_size>
 <output_size m:isArray="true" m:type="xs:int">256</output_size>
 <controllable_anatomy_size m:isArray="true">
  <anonymous_array m:isArray="true">hepatic tumor</anonymous_array>
  <anonymous_array m:isArray="true" m:type="xs:double">0.3</anonymous_array>
 </controllable_anatomy_size>
<controllable_anatomy_size m:isArray="true">
  <anonymous_array m:isArray="true">liver</anonymous_array>
  <anonymous_array m:isArray="true" m:type="xs:double">0.5</anonymous_array>
 </controllable_anatomy_size>
 <image_output_ext>.nii.gz</image_output_ext>
 <label_output_ext>.nii.gz</label_output_ext>
 <pre_signed_url></pre_signed_url>
 <spacing m:type="xs:double">1</spacing>
 <spacing m:type="xs:double">1</spacing>
 <spacing m:type="xs:double">1</spacing>
</root>
```

- **adept/fuyu-8b**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<messages m:isArray="true">
  <content>Hi! What is in this image?
    &lt;img src="
+wSzIAAAABlBMVEX///+/v7+jQ3Y5AAAADklEQVQI12P4AIX8EAgALgAD/
aNpbtEAAAAASUVORK5CYII=" /&gt;
  </content>
  <role>user</role>
</messages>
<temperature m:type="xs:double">0.2</temperature>
<top_p m:type="xs:double">0.7</top_p>
<max_tokens m:type="xs:int">1024</max_tokens>
<seed m:type="xs:int">42</seed>
<stream m:type="xs:boolean">false</stream>
</root>
```

- **Google/deplot**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<messages m:isArray="true">
<content>Generate underlying data table of the figure below:
&lt;img src="
+wSzIAAAABlBMVEX///+/v7+jQ3Y5AAAADklEQVQI12P4AIX8EAgALgAD/
aNpbtEAAAAASUVORK5CYII="/&gt;
</content>
<role>user</role>
</messages>
<temperature m:type="xs:double">0.2</temperature>
<top_p m:type="xs:double">0.7</top_p>
<max_tokens m:type="xs:int">1024</max_tokens>
<stream m:type="xs:boolean">false</stream>
</root>
```

- **Google/paligemma**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<messages m:isArray="true">
```

```
<content>Hi! What is in this image? &lt;img src="
v7+jQ3Y5AAAADklEQVQI12P4AIX8EAgALgAD/aNpbtEAAAAASUVORK5CYII="/&gt;
</content>
<role>user</role>
</messages>
<temperature m:type="xs:double">0.2</temperature>
<top_p m:type="xs:double">0.7</top_p>
<max_tokens m:type="xs:int">1024</max_tokens>
<stream m:type="xs:boolean">false</stream>
</root>
```

- **microsoft/kosmos-2**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<messages m:isArray="true">
<content>Where is the coach in this image? &lt;img src="
v7+jQ3Y5AAAADklEQVQI12P4AIX8EAgALgAD/aNpbtEAAAAASUVORK5CYII="/&gt;
</content>
<role>user</role>
</messages>
<grounded_response m:type="xs:boolean">true</grounded_response>
<temperature m:type="xs:double">0.2</temperature>
<top_p m:type="xs:double">0.7</top_p>
<max_tokens m:type="xs:int">1024</max_tokens>
<response_mode>brief</response_mode>
<task>VQA</task>
</root>
```

- **microsoft/phi-3-vision-128k-instruct**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<messages m:isArray="true">
<content>Hi! What is in this image? &lt;img src="
v7+jQ3Y5AAAADklEQVQI12P4AIX8EAgALgAD/aNpbtEAAAAASUVORK5CYII="/&gt;
</content>
<role>user</role>
</messages>
<temperature m:type="xs:double">0.2</temperature>
<top_p m:type="xs:double">0.7</top_p>
<max_tokens m:type="xs:int">1024</max_tokens>
<stream m:type="xs:boolean">false</stream>
</root>
```

- **briaai/bria-2.3**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<prompt>Give me image of an orange cat</prompt>
<aspect_ratio>1:1</aspect_ratio>
<mode>text-to-image</mode>
<negative_prompt></negative_prompt>
<model>bria-2.3</model>
<seed m:type="xs:int">0</seed>
<output_format>jpeg</output_format>
<cfg_scale m:type="xs:double">5</cfg_scale>
<steps m:type="xs:int">30</steps>
</root>
```

- **nvidia/nv-dinov2**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<messages m:isArray="true">
<content>
<type>image_url</type>
<image_url>
<url>
4gKwSUNDX1BST0ZJTEUAAQEAAAKgbGNtcwQwAABtbnRyUkdC
IFhZWiAH6AAIABUADQAcADdhY3NwQVBQTAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA9tYAAQAA
AADTLWxjbXMAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA1k
ZXNjAAABIAAAAEBjcHJ0AAABYAAAADZ3dHB0AAABmAAAABRjaGFkAAABrAAAACxyWFlaAAAB2AAA
```

```
ABRiWFlaAAAB7AAAABRnWFlaAAACAAAAABRyVFJDAAACFAAAACBnVFJDAAACFAAAACBiVFJDAAAC
FAAAACBjaHJtAAACNAAAACRkbW5kAAACWAAAACRkbWRkAAACfAAAACRtbHVjAAAAAAAAAEAAAAM
ZW5VUwAAACQAAAAcAEcASQBNAFAAIABiAHUAaQBsAHQALQBpAG4AIABzAFIARwBCbWx1YwAAAAAA
AAABAAAADGVuVVMAAAAaAAAAHABQAHUAYgBsAGkAYwAgAEQAbwBtAGEAaQBuAABYWVogAAAAAAAA
9tYAAQAAAADTLXNmMzIAAAAAAAEMQgAABd7///MlAAAHkwAA/ZD///uh///9ogAAA9wAAMBuWFla
IAAAAAAAAG+gAAA49QAAA5BYWVogAAAAAAAAJJ8AAA+EAAC2xFhZWiAAAAAAAABilwAAt4cAABjZ
cGFyYQAAAAAAAwAAAAJmZgAA8qcAAA1ZAAAT0AAAACltjaHJtAAAAAAADAAAAAKPXAABUfAAATM0A
AJmaAAAmZwAAD1xtbHVjAAAAAAAAAAAEAAAAMZW5VUwAAAAgAAAAcAEcASQBNAFBtbHVjAAAAAAAA
AAEAAAAMZW5VUwAAAAgAAAAcAHMAUgBHAEL/2wBDACgcHiMeGSgjISMtKygwPGRBPDc3PHtYXUlk kYCZlo
+AjIqgtObDoKrarYqMyP/L2u71////m8H////6/+b9//j/2wBDASstLTw1PHZBQXb4pYyl
+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj/wgARCACV
AOADAREAAhEBAxEB/8QAGQAAAwEBAQAAAAAAAAAAAAAAAAECAwQF/8QAFwEBAQEBAAAAAAAAAAAA
AAAAAAECA//aAAwDAQACEAMQAAAB5AEAAAwQAAAAAGADGMAKMVAAASjRZTMYABQASAxjLVxoIkZ
y0AAIzSWLNYoVONFmMaiywBQqA1NAEIRx0AA0uWBis6M3SgouWLJJJARqWMAEIRxBQMcMS53LXfO
puZqo3KlJZ1mwAsYCABCOFWBRUUYWTZcu0sWZ0ijpzdJc9Z2HBQAQUAKFXHNAyi4izn1AuXXOgLm
KzsuLlqOmx0DA5xjIADObY5KLObUhFVyuJ1KjfOmnNqZVYz0IYjM5EmkAAdPPox1UVZBz6jJSRVU
vTjXNvOdlmkirTOiySQEFgIR3cugKmNCsLMqDSXKzOy5QzsEZqXLZAgEIViEejz2oFLHQcOoHVm8
1CZ0E2CJQE1NpUIQhE2Ajt5dAZVjAzoIObUokkmwBGqBNC5ZFYhAIDpx0ocUWNCkclZ3LNpcLIpD
RqhDTQqVCsQgA6ufQplxSOgRFkkmFkUCBGJQQ02lRICoEm+dsqNCykBUACYVmZCqUYhzSsE6ZbJE
STYgmmtRpLpGiFSKxCGhWZz11RYznOahOyXQkkRJNmUorLl1l0hE1BlrMoUCEjKLzqpcN4o7JWSQ
STYjFWEUuudbZuWpnZNk2AgAAQUHKrmjplZJIrJEZqAMqa259MOnJBSQEAAADARaaqkBCEB//8QA
IhAAAgIBBQACAwAAAAAAAAAAAECESEQEiAwMRNBAyJA/9oACAEBAAEFAv6K6krNqHHltZgfNIoo
rolgsTy4Z2HxnxjUh7iiiiuESupey90vCwbxSssfrrStYldlZlwvRCYxxxFfrRRXc/NI+2SQ9Iui OSbpRl
+q87qJXrFlaYHE8PxvM/VKKE74zaFNG9G/O8+Rcpe6LhGBKI6jpVqMtrTtFoc0PlRXCaPr
bWtiefkJSb08FRKJ+N4yUUYMa50ri/No3Ztztxt0RL3VSKzwyZMlca0ofsVlIm83j7PvhGVFmTPL
BZZfBxTFGictIq2/eaY/4JaJtFpj4PgmWWWWX1uJsHAarqxzssvpnFs+NlD0+uEEmtiNiNhsNpt4
rjuN6N6N6JOL0htqolInDWCpdNi1sb6NzFJjbPSOGutSNw5s3MvsQiyy+hH13Lo//8QAHxEAAQME
AwEAAAAAAAAAAAAAAQARQBAgITAxUGBw/9oACAEDAQE/AfbHpAhHYUazEjiWFhEywEbRqf5029qP JFDG/
8QAHREAAgICAwEAAAAAAAAAAAAAAAAREAQBAwIVBgYf/aAAgBAgEBPwH2w0u2Z8rsxx4MAnNh
u2ZzALZg3rSO3HunuWFF2P8A/8QAIRAAAQMDBAMAAAAAAAAAAAAAAAMQAhQAERUBAgMGFBUWD/2gAI
AQEABj8CwZl21ba8eyfL+N/csyzGbg9bTqOJ68l6DQIISyiiji+5bTWwIQV6Siij8F//xAAnEAAD
AAIBBQEAAgEFAAAAAAAAAREQITEgQVFhcTCBkUChsdHh8f/aAAgBAQABPyH/ABoTCEJ+CXkDlbtW
YQgmcITR8dsYhsCtCBkUChsdHh8f/aAAgBAQABPyH/ABoTCEJ+CXkDlbtWwIQV6Siij8F//xAAnEAAD
0adyHHzTXlJfbPSOGutSNw5s3MvsQiyy+hH13Lo//8QAHxEAAQME
i1CE7mcmQexBoKKvts0+MIQkavsT8YJDZpBZhckn00YmN34O6duPPJRSn4QhBISKWo3VHiORV7qJ
T4aNBvC1c8PH14VCmjajOBO5bS/CCQhZbWh84g+LSBRd4NN3X+x4Rp8iEt6Yy1Yc80UmnRqk6Agh
OY59BqaNkIQSEM7o9kNmlwitVXCGCeOwjqqfZHc5C5XI6iEUWGrlol5GuLjZHkgkJCZZskYacn2H
IPbH4Etjjoac4U7Kyk228ktr/QbaTHZfkgTyPh4b8YR+cITPOMFmitGnRqvdjlN+MbsS69izFJae
0adyHHzTXljsfBDPI+iLyQhMIPRjvkUUam23yUC8R5CY3yHvZ3xSj0V2a9H2sEZhgi8kXknl0a1w
9og6IsOBIx7HycHPR2nwQaNGvwQsvgZtmzgDkxQMnxrKE6NJXhHwj4R8HwX0X0fwfwIXSlurQl7u
4TuXQidCcZRUaNdQgmJmupwnsa6XRGnwQuC2NTG94WU+T6Ps+ilExMYRSlIPZntvwxuvk+hnqE+E P
+FGI5nvhvDKXFKIOyvwxt+GV7GDbKUrym1wz2hnv/sRVv8Asdasqg2sPD6aJjipgewbFL+F1k2s
HgpS9LbGw3vDf6cilL0//9oADAMBAAIAAwAAABALbbJttt//ACWztpASTovTStO2tG2ykk/2bPD1
JkyJohIku6vKgIv7ttttZobrmYwgpNpNJJiw7ZS0AkpZtZJ5SQyASlg+7tJoEmAbsdDinSAtI5oC /Kql/
h0UZ1WWJtbu88AlI3pJkW1tnXgJ1aJMHIs2VNf5Wt6NUXA5Stptt/TpT6czgzqRNJdqYGK3
gFsu9+UFt5P9S52Ft0emStsy3NJztgg0eAyVeSN904AtqVMA2tp5butEgDF1uSRtApGuAG22wOMp JJ//
xAAeEQADAAEFAQEAAAAAAAAAAAAAAAAREQICEwMUBBUf/aAAgBAwEBPxD0Tka1TD8T2zCEEkRj
xCE50h470pj8nTC7J9Ho7Int4IQZcIr1P1zzLyhJD+hF8Y2WE4N3Y2KcFKUuJoeUN4SpEHGroT8D
y3naxpN2U09rivBBImKMmxeApSlKMWHj6Jj754QhCaOiiy/O8303yUvF8xNKVGphaYTVCEIQmh6b
cLhomI2yxspdSVHK5UxOkDaWDb4rtzNufBt8N3ifF//EAB4RAAMAAgMBAQEAAAAAAAAAAAAABERAg
ITAxQEFQ/9oACAECAQE/EP4TZWJ7XC+Jc4nBSlG2JoUzS97FjzMw1+i8+O5Muq4K9+Gi9JhkQsvF
8tfEWWNvAm/UJXhKiHKObu8IQhMXFwvBYYlcNj8EhNpzRrWbUpcp4tFnkhLwR1fD+Ci4KXkuGhcr 0g
+tMQhBeDZcIfg0LzvjIyaJtFErh5Wz+M1SC6GvhuF8MITpRSi0fAncMpSlLq8UpS9SYsPqaHok
JZmkG4KnO2DHIhAkuqc9zEuRYS6l1f/EACgQAQACAgICAQQCAgMAAAAAAAEAESExQVEQYZFxgaHR
ILEw8EDB8f/aAAgBAQABPxD/AACgQAQACAgICAQQCAgMAAAAAAAEAESExQVEQYZFxgaHRZxCT+Woz2BE4N3Y2KcFKUuJoeUN4SpEHGroT8D
HX3KOJULQgFwDpOEiYZFxgaHRZxCT+WoZ2BE4N3Y2KcFKUuJoeUN4SpEHGroT8D
2xglGyoSnLqCoIODuZggydRoWwBjzsEIUlkqVEiSpUBOKvE6AgwS4ib3CcSBVMXvAhKeH90Awv1y
ib096+Y5QSZeGG/cUbnGeIuE2w43L1U4YIpgAoleLAW2SkPEdGSPQAxUscwHIr6IeA8EpGZljxC5
VAz0QqwzMUZxjmF3nMNvDEN2h6gMsoziYSbIGn3FVK2sL5nA79wAjXJCWp8aBfeEQWfRBTnEU6lu
Jayl3PVDwHmDN7TE5jiEDgbHbGNwu7iqmjZLdIwMICC2XhRcC2PhIc6ItAdDXxKARq00+pLFL4h3
r9484PvEtiV8n7T0SWcRfEkRdEeTwfR4AwINCGJRrB4CX1PQdEQ0BdBCxbrL7iKNKpKiHuDA4P7h
MHIqoLTD3BqEP8R/UCbXhxxHktvNETikxx8pfEkfUfaPZUeaFc/nO4R94+5PqIexD3IyQmo1WbvU
SiXTcA87a1EatGCKsqtUsUaDAShxd7Ygw03OL94FUTfcbGEJU5WyWAAvKrCfuP8AqInXHn/KL5/K J6/
Mev8AM9fwEdQHUA8QPUp1L9SwBHsZqK/cUX32K25uYBo2ykG11/cvelzmg2si2Vl2ZiVBhmek
4emJuqTdGGPt+I+34j7fiY7/ABMdzHcx3CFQSLwGXGqrqcKo0RFmO5Y67gcAZkeY00RwdTiolMVN
pfqIZaFzRxBszNNS4jWhTuf+RLyVJ9E+hPpT6EqCKdSnUK6mOpjqY6jJLMM23pG3R9MeA0vEB1zM v0iQW
+paq43K6iWX14qGFGmPDF7Su0amO5iEHpA6h9QniAoh4uXNwwPaOK2miB056ZSFEt8TgS+v
```

```
UrGIbVdPEZ5KhENcjHqfmcFY8KiO3xH/AGIs4Qg8/RixGGUm2I7ZTB2PxCvbNqR5qNciUEU5LIbz
44OsKchAtH6/pGuHiC2LjZV1FOoHTF9Reo+kfSX4CI04Q/EKf+iB/RHpipm5e5ipaYqtlszNsH0Y H+/
wApJIdgt7iAAfqRELFn2Rv1LRuMuXCFINASVWUl6rqLco/axbz4XLly5cuCxWNwWJszKQXH2n
sj7RlhX8BlAl+hGYuI/4nR41QVRUtFiy5c//2Q==</url>
</image_url>
</content>
</messages>
</root>
```


- **nvidia/nv-grounding-dino**


```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<model>Grounding Dino</model>
<messages m:isArray="true">
<role>user</role>
<content m:isArray="true">
<type>text</type>
<text>person, cars, bus, backpack</text>
</content>
<content m:isArray="true">
<type>media_url</type>
<media_url>
<url>
2wBDAAcFBQYFBAcGBQYIBwcIChELCgkJChUPEAwRGBUaGRgVGBcbHichGx0lHRcYIi4iJSgpKywrGiAvMy8qM
icqKyr/
2wBDAQcICAoJChQLCxQqHBgcKioqKioqKioqKioqKioqKioqKioqKioqKioqKioqKioqKioqK
ioqKir/wAARCABAAHIDASIAAhEBAxEB/8QAHAAAAgIDAQEAAAAAAAAAAAAABAUDBgABAgcI/
8QANBAAAgEDAwIEBQQBAwUAAAAAAQIDAAQRBRIhMUEGE1FhFCJxgaEykbHRwQcj4RUmQmLx/
8QAGgEAAgMBAQAAAAAAAAAAAAAAAAAAwQBAgUABv/
EACgRAAIBAwMCBgMBAAAAAAAAAAAABAgMRIQQSMSJBEyNRcaHwFDJhwf/
aAAwDAQACEQMRAD8AIW29q7Fv7U3yOMgcxOMgcvC2LL2r02480qNyOFIgqRIPaj
```

```
j6gBskew5q/CK9y0G9AJD8sOp96yo0s96BmC5IyeaykegaP/2Q==</url>
</media_url>
</content>
</messages>
<threshold m:type="xs:double">0.7</threshold>
</root>
```

- **stabilityai/stable-diffusion-3-medium**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<aspect_ratio>1:1</aspect_ratio>
<cfg_scale m:type="xs:double">5</cfg_scale>
<mode>text-to-image</mode>
<model>sd3</model>
<negative_prompt></negative_prompt>
<output_format>jpeg</output_format>
<prompt>An orange cat</prompt>
<seed m:type="xs:int">0</seed>
<steps m:type="xs:int">50</steps>
</root>
```

- **stabilityai/sdxl-turbo**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<height m:type="xs:int">512</height>
<width m:type="xs:int">512</width>
<text_prompts m:isArray="true">
<text>A photo of a Shiba Inu dog with a backpack riding a bike</text>
<weight m:type="xs:double">1</weight>
</text_prompts>
<cfg_scale m:type="xs:double">0</cfg_scale>
<clip_guidance_preset>NONE</clip_guidance_preset>
<sampler>K_EULER_ANCESTRAL</sampler>
<samples m:type="xs:int">1</samples>
<seed m:type="xs:int">0</seed>
<steps m:type="xs:int">4</steps>
<style_preset>none</style_preset>
</root>
```

- **stabilityai/stable-diffusion-xl**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<height m:type="xs:int">1024</height>
<width m:type="xs:int">1024</width>
<text_prompts m:isArray="true">
<text>A photo of a Shiba Inu dog with a backpack riding a bike</text>
<weight m:type="xs:double">1</weight>
</text_prompts>
<cfg_scale m:type="xs:double">5</cfg_scale>
<clip_guidance_preset>NONE</clip_guidance_preset>
<sampler>K_DPM_2_ANCESTRAL</sampler>
<samples m:type="xs:int">1</samples>
<seed m:type="xs:int">0</seed>
<steps m:type="xs:int">25</steps>
<style_preset>none</style_preset>
</root>
```

- **stabilityai/stable-video-diffusion**

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<image>
8DTgE8HuxwEQAAAABJRU5ErkJggg==</image>
<seed m:type="xs:int">0</seed>
<cfg_scale m:type="xs:double">1.8</cfg_scale>
<motion_bucket_id m:type="xs:int">127</motion_bucket_id>
</root>
```

- The **Invoke Climate Simulation Model** action invokes the Climate Simulation API models, such as nvidia/fourcastnet. It runs models for climate prediction and environmental simulation.
  The connector is tested with the Nvidia/fourcastnet model.

  You must provide values to the input fields, such as payload, version, and NVCF_POLL_SECONDS.

For example, consider the nvidia/fourcastnet model with the following endpoint URL:

```
https://climate.api.nvidia.com/v1/nvidia/fourcastnet
```

In this URL, `v1` is the API version and `nvidia`, and `fourcastnet` are individual URL path segments.

The following snippet is a payload example:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<input_id m:type="xs:int">0</input_id>
<variables>w10m,z500</variables>
<simulation_length m:type="xs:int">4</simulation_length>
<ensemble_size m:type="xs:int">1</ensemble_size>
<noise_amplitude m:type="xs:double">0</noise_amplitude>
</root>
```

The following snippet is a sample response:

```
200 OK
fourcastnet.zip
```

- The **Invoke Medical Imaging Model** action invokes the Healthcare Medical Imaging API models, such as nvidia/vista-3d and nvidia/maisi. It invokes models for analyzing medical images, such as scans or X-rays. The connector is tested with the following models:

  - nvidia/vista-3d

  - nvidia/maisi

  You must provide values to the input fields, such as payload, version, and NVCF_POLL_SECONDS.

  For example, consider the nvidia/vista-3d model with the following endpoint URL:

  ```
  https://health.api.nvidia.com/v1/medicalimaging/nvidia/vista-3d
  ```

  In this URL, `v1` is the API version and `medicalimaging`, `nvidia`, and `vista-3d` are individual URL path segments.

  The following snippet is a payload example:

  ```
  <root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
  <image>https://assets.ngc.nvidia.com/products/api-catalog/vista3d/example-1.nii.gz</
  image>
  <prompts>
  <classes m:isArray="true">spleen</classes>
  <points>
  <spleen m:isArray="true">
  <anonymous_array m:isArray="true" m:type="xs:int">97</anonymous_array>
  <anonymous_array m:isArray="true" m:type="xs:int">249</anonymous_array>
  <anonymous_array m:isArray="true" m:type="xs:int">87</anonymous_array>
  </spleen>
  </points>
  </prompts>
  </root>
  ```

  This XML structure uses metadata attributes such as `m:isArray` and `m:type` to represent JSON-like arrays and types. It also leverages custom fields named `anonymous_array` to support anonymous arrays within the payload structure.

  The following snippet is an equivalent JSON representation:

  ```
  {
   "image": "https://assets.ngc.nvidia.com/products/api-catalog/vista3d/
  example-1.nii.gz",
   "prompts": {
    "classes": [
     "spleen"
    ],
    "points": {
     "spleen": [
      [
       97,
       249,
       87
  ```

```
      ]
     ]
    }
   }
}
```

The following snippet is a sample response:

```
200 OK
vista-3d.zip
```

- The **Invoke Multimodal Model** action invokes the Multimodal and Visual API models, such as nvidia/neva-22b and nvidia/nv-grounding-dino. It invokes models that process and combine different data types, such as text, images, or video.
  The connector is tested with the following models:

  - adept/fuyu-8b
  - Google/deplot
  - google/paligemma
  - microsoft/kosmos-2
  - microsoft/phi-3-vision-128k-instruct
  - nvidia/neva-22b
  - briaai/bria-2.3
  - nvidia/nv-dinov2
  - nvidia/nv-grounding-dino
  - stabilityai/stable-diffusion-3-medium
  - stabilityai/sdxl-turbo
  - stabilityai/stable-diffusion-xl
  - stabilityai/stable-video-diffusion

  You must provide values to the input fields, such as payload, version, and NVCF_POLL_SECONDS.

  For example, consider the nvidia/neva-22b model with the following endpoint URL:

  ```
  https://ai.api.nvidia.com/v1/vlm/nvidia/neva-22b
  ```

  In this URL, `v1` is the API version and `vlm`, `nvidia`, and `neva-22b` are individual URL path segments.

  The following snippet is a payload example:

  ```
  <root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
  <messages m:isArray="true">
  <content>Hi! What is in this image? &lt;img src="
  4AAQSkZJRgABAQEBLAEsAAD/4gKwSUNDX1BST0ZJTEUAAQEAAAKgbGNtcwQwAABtbnRyUkdC
  IFhZWiAH6AAIABUADQAcADdhY3NwQVBQTAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA9tYAAQAA
  AADTLWxjbXMAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA1k
  ZXNjAAABIAAAAEBjcHJ0AAABYAAAADZ3dHB0AAABmAAAABRjaGFkAAABrAAAACxyWFlaAAAB2AAA
  ABRiWFlaAAAB7AAAABRnWFlaAAACAAAAABRyVFJDAAACFAAAACBnVFJDAAACFAAAACBiVFJDAAAC
  FAAAACBjaHJtAAACNAAAACRkbW5kAAACWAAAACRkbWRkAAACfAAAACRtbHVjAAAAAAAAAAEAAAAM
  ZW5VUwAAACQAAAAcAEcASQBNAFAAIABiAHUAaQBsAHQALQBpAG4AIABzAFIARwBCbWx1YwAAAAAA
  AAABAAAADGVuVVMAAAAaAAAAHABQAHUAYgBsAGkAYwAgAEQAbwBtAGEAaQBuAABYWVogAAAAAAAA
  9tYAAQAAAADTLXNmMzIAAAAAAAEMQgAABd7///MlAAAHkwAA/ZD///uh///9ogAAA9wAAMBuWFla
  IAAAAAAAAG+gAAA49QAAA5BYWVogAAAAAAAAJ8AAA+EAAC2xFhZWiAAAAAAAABilwAAt4cAABjZ
  cGFyYQAAAAAAAwAAAAJmZgAA8qcAAA1ZAAAT0AAACltjaHJtAAAAAAADAAAAAKPXAABUfAAATM0A
  AJmaAAAmZwAAD1xtbHVjAAAAAAAAAAEAAAAMZW5VUwAAAAgAAAAcAEcASQBNAFBtbHVjAAAAAAAA
  AAEAAAAMZW5VUwAAAAgAAAAcAHMAUgBHAEL/2wBDACgcHiMeGSgjISMtKygwPGRBPDc3PHtYXUlk kYCZlo
  +AjIqgtObDoKrarYqMyP/L2u71////m8H////6/+b9//j/2wBDASstLTw1PHZBQXb4pYyl
  +Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj4+Pj/wgARCACV
  AOADAREAAhEBAxEB/8QAGQAAAwEBAQAAAAAAAAAAAAAAAAECAwQF/8QAFwEBAQEBAAAAAAAAAAAA
  AAAAAAECA//aAAwDAQACEAMQAAAB5AEAAAwQAAAAAGADGMAKMVAAASjRZTMYABQASAxjLVxoIkZ
  y0AAIzSWLNYoVONFmMaiywBQqA1NAEIRx0AA0uWBis6M3SgouWLJJJARqWMAEIRxBQMcMS53LXfO
  puZqo3KlJZ1mwAsYCABCOFWBRUUYWTZcu0sWZ0ijpzdJc9Z2HBQAQUAKFXHNAyi4izn1AuXXOgLm
  KzsuLlqOmx0DA5xjIADObY5KLObUhFVyuJ1KjfOmnNqZVYz0IYjM5EmkAAdPPox1UVZBz6jJSRVU
  ```

```
vTjXNvOdlmkirTOiySQEFgIR3cugKmNCsLMqDSXKzOy5QzsEZqXLZAgEIViEejz2oFLHQcOoHVm8
1CZ0E2CJQE1NpUIQhE2Ajt5dAZVjAzoIObUokkmwBGqBNC5ZFYhAIDpx0ocUWNCkclZ3LNpcLIpD
RqhDTQqVCsQgA6ufQplxSOgRFkkmFkUCBGJQQ02lRICoEm+dsqNCykBUACYVmZCqUYhzSsE6ZbJE
STYgmmtRpLpGiFSKxCGhWZz11RYznOahOyXQkkRJNmUorLl1l0hE1BlrMoUCEjKLzqpcN4o7JWSQ
STYjFWEUuudbZuWpnZNk2AgAAQUHKrmjplZJIrJEZqAMqa259MOnJBSQEAAADARaaqkBCEB//8QA
IhAAAgIBBQACAwAAAAAAAAAAAECESEQEiAwMRNBAyJA/9oACAEBAAEFAv6K6krNqHHltZgfNIoo
rolgsTy4Z2HxnxjUh7iiiiuESupey90vCwbxSssfrrStYldlZlwvRCYxxxFfrRRXc/NI+2SQ9Iui OSbpRl
+q87qJXrFlaYHE8PxvM/VKKE74zaFNG9G/O8+Rcpe6LhGBKI6jpVqMtrTtFoc0PlRXCaPr
bWtiefkJSb08FRKJ+N4yUUYMa50ri/No3Ztztxt0RL3VSKzwyZMlca0ofsVlIm83j7PvhGVFmTPL
BZZfBxTFGictIq2/eaY/4JaJtFpj4PgmWWWWX1uJsHAarqxzssvpnFs+NlD0+uEEmtiNiNhsNpt4
rjuN6N6N6JOL0htqolInDWCpdNi1sb6NzFJjbPSOGutSNw5s3MvsQiyy+hH13Lo//8QAHxEAAQME
AwEAAAAAAAAAAAAAAAQARQBAgITAxUGBw/9oACAEDAQE/AfbHpAhHYUazEjiWFhEywEbRqf5029qP JFDG/
8QAHREAAgICAwEAAAAAAAAAAAAAAAREAQBAwIVBgYf/aAAgBAgEBPwH2w0u2Z8rsxx4MAnNh
u2ZzALZg3rSO3HunuWFF2P8A/8QAIRAAAQMDBAMAAAAAAAAAAAAAAAMQAhQAERUBAgMGFBUWD/2gAI
AQEABj8CwZl21ba8eyfL+N/csyzGbg9bTqOJ68l6DQIISyiiji+5bTWwIQV6Siij8F//xAAnEAAD
AAIBBQEAAgEFAAAAAAAAAREQITEgQVFhcTCBkUChsdHh8f/aAAgBAQABPyH/ABoTCEJ+CXkD1tbW
YQgmcITR8EdswhBcFGaLBhLqg2idsPOHcWivX9iZ/wDRH/onpLXoo9GU+gSCRIIaEIQnUqhsahM7
i1CE7mcmQexBoKKvts0+MIQkavsT8YJDZpBZhckn00YmN34O6duPPJRSn4QhBISKWo3VHiORV7qJ
T4aNBvC1c8PH14VCmjajOBO5bS/CCQhZbWh84g+LSBRd4NN3X+x4Rp8iEt6Yy1Yc80UmnRqk6Agh
OY59BqaNkIQSEM7o9kNmlwitVXCGCeOwjqqfZHc5C5XI6iEUWGrlol5GuLjZHkgkJCZZskYacn2H
IPbH4Etjjoac4U7Kyk228ktr/QbaTHZfkgTyPh4b8YR+cITPOMFmitGnRqvdjlN+MbsS69izFJae
0adyHHzTXljsfBDPI+iLyQhMIPRjvkUUam23yUC8R5CY3yHvZ3xSj0V2a9H2sEZhgi8kXknl0alw
9og6IsOBIx7HycHPR2nwQaNGvwQsvgZtmzgDkxQMnxrKE6NJXhHwj4R8HwX0X0fwfwIXSlurQl7u
4TuXQidCcZRUaNdQgmJmupwnsa6XRGnwQuC2NTG94WU+T6Ps+ilExMYRSlIPZntvwxuvk+hnqE+E P
+FGI5nvhvDKXFKIOyvwxt+GV7GDbKUrym1wz2hnv/sRVv8Asdasqg2sPD6aJjipgewbFL+F1k2s
HgpS9LbGw3vDf6cilL0//9oADAMBAAIAAwAAABALbbJttt//ACWztpASTovTStO2tG2ykk/2bPD1
JkyJohIku6vKgIv7ttttZobrmYwgpNpNJJiw7ZS0AkpZtZJ5SQyASlg+7tJoEmAbsdDinSAtI5oC /Kql/
h0UZ1WWJtbu88AlI3pJkW1tnXgJ1aJMHIs2VNf5Wt6NUXA5Stptt/TpT6czgzqRNJdqYGK3
gFsu9+UFt5P9S52Ft0emStsy3NJztgg0eAyVeSN904AtqVMA2tp5butEgDF1uSRtApGuAG22wOMp JJ//
xAAeEQADAAEFAQEAAAAAAAAAAAAAAAREQICEwMUBBUf/aAAgBAwEBPxD0Tka1TD8T2zCEEkRj
xCE50h470pj8nTC7J9Ho7Int4IQZcIr1P1zzLyhJD+hF8Y2WE4N3Y2KcFKUuJoeUN4SpEHGroT8D
y3naxpN2U09rivBBImKMmxeApSlKMWHj6Jj754QhCaOiiy/O8303yUvF8xNKVGphaYTVCEIQmh6b
cLhomI2yxspdSVHK5UxOkDaWDb4rtzNufBt8N3ifF//EAB4RAAMAAgMBAQEAAAAAAAAAAAAABERAg ITAxQEFQ/
9oACAECAQE/EP4TZWJ7XC+Jc4nBSlG2JoUzS97FjzMw1+i8+O5Muq4K9+Gi9JhkQsvF 8tfEWWNvAm/
UJXhKiHKObu8IQhMXFwvBYYlcNj8EhNpzRrWbUpcp4tFnkhLwR1fD+Ci4KXkuGhcr 0g
+tMQhBeDZcIfg0LzvjIyaJtFErh5Wz+M1SC6GvhuF8MITpRSi0fAncMpSlLq8UpS9SYsPqaHok
JZmkG4KnO2DHIhAkuqc9zEuRYS6l1f/EACgQAQACAgICAQQCAgMAAAAAAAEAESExQVEQYZFxgaHR
ILEw8EDB8f/aAAgBAQABPxD/AIdSoEvCRdRZxCT+Woz2PdKzlqxP9CAq1NNcSpUylvAm1rtwQgOLLL
HX3KOJUYlaWFWPDGlaWpwPEQwGHLL3Ea1ZtJCTFrh3WZP4VBqXWN/b3LXN9oYs7lressrid6YzSle1wpEVeoLpw1BNaxU
Ear24hjprvvxJIWkyUktaTEDpGNowxXioEC4RGi8e2Iq7lRs+YW12vUekqt6hcrPqLpCqaJsK6cQ
mXH9Q0S9EQbzGPhc4KUGe4laLgDRKlRIkSVKlQ8A9PG5cNfbw2kGRGl+CBrykQct/ucp+6aFk5SK
3DeIykX2RiHlWZeA8wdCVKlSpUSJ4WhGSDDfYZ+87EFpleyEBpR3HUwKaYIaqb3LKthpjL6SLiF3
2xglGyoSnLqCoIODuZggydRoWwBjzsEIUlkqVEiSpUBOKvE6AgwS4ib3CcSBVMXvAhKeH90Awv1y
ib096+Y5QSZeGG/cUbnGeIuE2w43L1U4YIpgAoleLAW2SkPEdGSPQAxUscwHIr6IeA8EpGZljxC5
VAz0QqwzMUZxjmF3nMNvDEN2h6gMsoziYSbIGn3FVK2sL5nA79wAjXJCWp8aBfeEQWfRBTnEU6lu
Jayl3PVDwHmDN7TE5jiEDgbHbGNwu7iqmjZLdIwMICC2XhRcC2PhIc6ItAdDXxKARq00+pLFL4h3
r9484PvEtiV8n7T0SWcRfEkRdEeTwfR4AwINCGJRrB4CX1PQdEQ0BdBCxbrL7iKNKpKiHuDA4P7h
MHIqoLTD3BqEP8R/UCbXhxxHktvNETikxx8pfEkfUfaPZUeaFc/nO4R94+5PqIexD3IyQmo1WbvU
SiXTcA87a1EatGCKsqtUsUaDAShxd7Ygw03OL94FUTfcbGEJU5WyWAAvKrCfuP8AqInXHn/KL5/K J6/
Mev8AM9fwEdQHUA8QPUp1L9SwBHsZqK/cUX32K25uYBo2ykG11/cvelzmg2si2Vl2ZiVBhmek 4emJuqTdGGPt
+I+34j7fiY7/ABMdzHcx3CFQSLwGXGqrqcKo0RFmO5Y67gcAZkeY00RwdTiolMVN
pfqIZaFzRxBszNNS4jWhTuf+RLyVJ9E+hPpT6EqCKdSnUK6mOpjqY6jJLMM23pG3R9MeA0vEB1zM v0iQW
+paq43K6iWX14qGFGmPDF7Su0amO5iEHpA6h9QniAoh4uXNwwPaOK2miB056ZSFEt8TgS+v
UrGIbVdPEZ5KhENcjHqfmcFY8KiO3xH/AGIs4Qg8/RixGGUm2I7ZTB2PxCvbNqR5qNciUEU5LIbz
44OsKchAtH6/pGuHiC2LjZV1FOoHTF9Reo+kfSX4CI04Q/EKf+iB/RHpipm5e5ipaYqtlszNsH0Y H+/
wApJIdgt7iAAfqRELFn2Rv1LRuMuXCFINASVWUl6rqLco/axbz4XLly5cuCxWNwWJszKQXH2n sj7RlhX8BlAl
+hGYuI/4nR41QVRUtFiy5c//2Q==
```


"/>
</content>
<name>John</name>
<role>user</role>
</messages>
<temperature m:type="xs:double">0.2</temperature>
<top_p m:type="xs:double">0.7</top_p>
<max_tokens m:type="xs:int">1024</max_tokens>
<seed m:type="xs:int">42</seed>
<stream m:type="xs:boolean">false</stream>
</root>

The following snippet is a sample response:

```
<root>
  <usage>
   <completion_tokens>37</completion_tokens>
   <prompt_tokens>754</prompt_tokens>
   <total_tokens>791</total_tokens>
  </usage>
  <id>cd05322e-71e5-4b01-a76f-153d96b7a8c0</id>
  <choices>
   <finish_reason>stop</finish_reason>
   <index>0</index>
   <message>
     <role>assistant</role>
     <name/>
     <content>This image shows a motorcycle on a racetrack. The motorcycle is a
Kawasaki Ninja H2R. The motorcycle is a super sport motorcycle. It is a track-only
motorcycle.</content>
   </message>
  </choices>
</root>
```

- The **Invoke Basic Model** action invokes the basic models for chat completions, function calling, text reranking, and structured output actions. It executes actions, such as chat completions, function calling, embeddings, text reranking, and structured output actions.
  You must provide values to the input fields, such as host, version, segments, and payload.

  The following snippet is an example of executing Chat completion action payload:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <model>mistralai/mixtral-8x7b-instruct-v0.1</model>
    <max_tokens m:type="xs:int">1024</max_tokens>
    <stream m:type="xs:boolean">false</stream>
    <temperature m:type="xs:double">1.2</temperature>
    <top_p m:type="xs:double">0.8</top_p>
    <frequency_penalty m:type="xs:double">0</frequency_penalty>
    <presence_penalty m:type="xs:double">0</presence_penalty>
    <seed m:type="xs:int">1024</seed>
    <messages m:isArray="true">
        <role>user</role>
        <content>You are a helpful assistant.</content>
    </messages>
    <messages m:isArray="true">
        <role>assistant</role>
        <content>You are a helpful assistant because you help people with their
problems and you are always there for them when they need you.</content>
    </messages>
    <messages m:isArray="true">
        <role>user</role>
        <content>Who is Leo Messi.</content>
    </messages>
</root>
```

  The following snippet is an example to execute Chat Completion Function Call action payload:

```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <model>meta/llama-3.1-70b-instruct</model>
    <messages m:isArray="true">
        <role>user</role>
        <content>My name is John and I'm 30 years old</content>
    </messages>
    <tools m:isArray="true">
        <type>function</type>
        <function>
            <name>get_name_age</name>
            <description>Please provide your name and age</description>
            <parameters>
                <type>object</type>
                <properties>
                    <name>
                        <type>string</type>
                        <description>The name of participant</description>
```

```
                    </name>
                    <age>
                        <type>integer</type>
                        <description>The age of participant</description>
                    </age>
                </properties>
                <required m:isArray="true">name</required>
                <required m:isArray="true">age</required>
            </parameters>
        </function>
    </tools>
    <tool_choice>
        <type>function</type>
        <function>
            <name>get_name_age</name>
        </function>
    </tool_choice>
</root>
```

The following snippet is an example to execute Embeddings action payload:

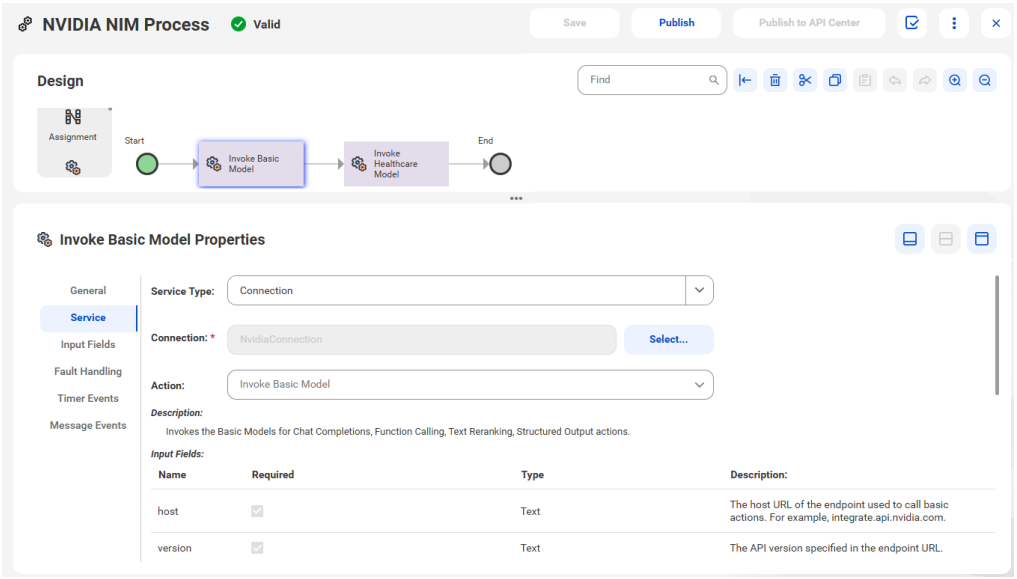
```
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
    <input m:isArray="true">Paris is the capital of France</input>
    <model>nvidia/nv-embedqa-e5-v5</model>
    <input_type>query</input_type>
</root>
```

Note: Reranking and Structured Output actions can be executed on local environment, Docker image, or on-premise hosting environment.

Using an NVIDIA NIM connection in a process

After you create an NVIDIA NIM connection, you can use it in a Service step in a process.

1. Add a Service step to the process.
2. Click the **Service** tab.
3. From the **Service Type** list, select **Connection**.
4. From the **Connection** list, browse and select the NVIDIA NIM connection that you created.
5. From the **Action** list, select the action that you want to perform. For example, select the **Invoke Basic Model** action to invoke the basic models for chat completions, function calling, text reranking, and structured output actions as shown in the following image:



In this case, Application Integration populates the **host**, **version**, **segment1**, …, **segment5**, and **payload** objects under the input fields and the **response** and **statusCode** objects under the output fields.

6. Configure further steps as needed.
7. Validate, save, and publish the process.